

SAI SAMPATH KEDARI

✉ sampath@umich.edu ☎ 734-510-1994 🌐 [saisampathkedari](https://saisampathkedari.com) 🌐 [SaiSampathKedari](https://SaiSampathKedari.com) in [sai-sampath-kedari](https://sai-sampath-kedari.github.io) 🐦

WORK EXPERIENCE

- Abundance Intelligence:** Founding Researcher | *San Francisco, CA* **Feb 2026 – May 2026**
- Developed a reward-learning framework that converts egocentric human demonstration videos into dense reward signals for training and evaluating robotic manipulation policies.
- Peer Robotics:** Motion and Controls Intern | *Detroit, MI* **Jul 2025 – Sep 2025**
- Developed encoder-based human-force detection for manual takeover and ROS 2 motion-planning logic for an autonomous pallet jack, accounting for passive-trailer kinematics to guide the uninstrumented trailer.
- iRaL, University of Michigan:** Graduate Researcher | *Ann Arbor, MI* **Aug 2024 – Jun 2025**
- Modeled nonlinear residual dynamics in high-speed quadrotor flight using Gaussian processes and learned Koopman operators to improve the dynamics-model fidelity required for precise feedback control at high speeds.
 - Developed Bayesian state-estimation methods under process and measurement uncertainty to provide the accurate robot-state information required for reliable high-speed feedback control.
- Dassault Systèmes & Altair Engineering:** R&D Software Developer | *India* **Aug 2019 – Aug 2021**
- Developed C++ features and APIs for CATIA FTA and HyperMesh, integrating Python and Tcl/Tk.

OPEN-SOURCE RESEARCH SOFTWARE

- Bimanual Robot Learning with OpenArm** | *ROS 2, MuJoCo* **In Development**
- Building an end-to-end bimanual manipulation pipeline for a dual-arm 14-DOF OpenArm, integrating MuJoCo simulation, ROS 2 control, and LeRobot-format demonstration collection toward behavior cloning, ACT, and diffusion-policy evaluation on long-horizon tableware tasks.
- Deep RL Algorithms Library** | [GitHub](#) | [Derivations](#) | [Experiments](#) **In Development**
- Developing a unified PyTorch deep-RL library covering on-policy (REINFORCE, A2C, NPG, TRPO, PPO) and off-policy (DQN, DDPG, TD3, SAC) methods, implementing GAE, Fisher-vector products with conjugate-gradient natural gradients, KL trust-region line search, and clipped surrogate objectives from first principles.
 - Benchmarking on MuJoCo locomotion suites (HalfCheetah, Hopper, Walker2d) with multi-seed runs, analyzing sample efficiency, estimator bias-variance, and training stability across algorithm families.
- Reinforcement Learning Algorithms Library** | [GitHub](#) | [Derivations](#) | [Experiments](#)
- Turned Sutton & Barto into an executable theory-to-code reference, pairing modular NumPy implementations of DP, Monte Carlo, TD, eligibility traces, SARSA, Q-learning, n -step and semi-gradient methods, state aggregation, and tile coding with 17 first-principles derivations extending through policy gradients and GAE.
 - Reproduced the book's published benchmark results across Blackjack, Windy Gridworld, Random Walk, and Mountain Car, matching known learning curves in both tabular and continuous-state settings.
- Bayesian State & Parameter Estimation** | [State Estimation](#) | [Parameter Inference](#) | [Derivations](#)
- Built a Bayesian estimation library for robot state estimation and sensor fusion under nonlinear dynamics: KF, EKF, UKF, Gauss-Hermite, and particle filters from scratch, benchmarked on nonlinear pendulum tracking, with derived optimal proposals, ESS-based resampling, and MCMC-based parameter identifiability analysis.
- Monte Carlo Sampling & Inference Library** | [GitHub](#) | [Derivations](#) | [Experiments](#)
- Built a Monte Carlo library for computational inference: importance sampling (standard and self-normalized), variance reduction with control variates, and adaptive MCMC (MH, Adaptive Metropolis, Delayed Rejection, DRAM); validated on rare-event estimation and banana-shaped posteriors with IAC and ESS diagnostics.

EDUCATION

- University of Michigan, Ann Arbor** **Aug 2021 – Apr 2024**
M.S. in Mechanical Engineering – Robotics, Mathematics & Statistics
M.S. in Automotive Engineering – Control & Machine Learning
- National Institute of Technology Rourkela, India** **Jul 2015 – May 2019**
B.Tech. in Mechanical Engineering

TECHNICAL SKILLS

Languages & Tools: C++, Python, PyTorch, NumPy, MuJoCo, Gymnasium, ROS 2, SciPy